Behavior-Oriented Simulation

Project: Food Waste Reduction Platform

Delivery: Workshop 4 - System Simulation

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Date: May 2026

Version: 1.0 (Collaborative Draft)

Purpose

The main objective of the **Behavior-Oriented Simulation** is to model and analyze the **dynamic interactions, decisions, and emergent behaviors** of the key actors:

1. Donors
2. Cafeterias
3. Individual beneficiaries
4. Verified organizations
5. Volunteers

Within the food surplus redistribution system.

This simulation complements the **Process-Oriented Simulation**, which focuses on evaluating how the system flows at a macro level (matching latency, throughput). The Behavior-Oriented Simulation deepens into why and how individual and collective behaviors evolve, capturing phenomena such as:

- No-shows and their cascading propagation.
- Effects of staged notifications on fairness and demand saturation.
- Impact of reliability scoring on participation over time.
- User adoption and disengagement dynamics.

Linked Project Success Criteria

- Achieve ≥ 80% surplus recovery rate.

- ≥ 85% pickup completion rate.
- Fairness in allocation (20%+ priority for charity).
- Mitigation of the 10 critical risks identified in Workshop 1.

This simulation seeks to provide quantitative and qualitative evidence that validates the risk mitigation mechanisms and architectural patterns defined in Workshop 2 and redefined in Workshop 3.

The results directly feed the implementation of the functional prototype (Workshop 4 / Final Delivery).

Simulation Model Overview

Simulation Type

Behavior-oriented (Agent-Based Modeling ABM) with rule-based simulation elements and simple adaptive learning.

Modeling Paradigm

Multi-agent.

Each actor (donor, individual beneficiary, charity, volunteer) is an autonomous agent with attributes, decision rules, and internal state that evolves over time.

Tool to Use

Python Mesa

Temporal Representation

Discrete time with simulation steps of 5-15 minutes (within the operational windows of 2-4 publication hours and 1-3 collection hours).

Support for asynchronous events.

Spatial Resolution

Simplified geo-referenced model of the 3 km radius around the Faculty of Engineering (UD).

Use of cartesian coordinates or grid to calculate distances.

Key Focus

Individual and emergent behaviors: probabilistic decision-making, simple reinforcement learning (reliability update), social interaction, and response to notifications.

Central Implemented Mechanism

- Serialized priority matching algorithm with composite scoring (proximity + reliability + charity priority)
- Stochastic no-show rules that affect future scoring.
- Staged notifications by radius (500m → 1km → 3km) with variable delays.

System Boundaries and Assumptions

Boundaries

Inside the Model

1. Autonomous behavior of agents.
2. Complete surplus publication process.
3. Serialized matching algorithm with multi-criteria scoring (proximity, reliability, and charity priority)
4. Staged notification system (by radius and with safeguards)
5. Probabilistic decision-making of agents (accept/reject offer)
6. No-show mechanism and its impact on the user's reliability scoring.
7. Automatic reassignment of uncollected surpluses.
8. Temporal evolution of reliability scoring and participation level of agents.
9. Fairness, efficiency, and surplus recovery metrics.
10. Emergent dynamics (demand saturation, no-show cascade effect, adaptive learning of users)

Outside the Model

1. Physical transportation of food from donor to beneficiary.

2. Verification of quality, condition, and food safety of surpluses.
3. Payment or economic compensation processes between parties.
4. Internal inventory management, preparation, and food storage in cafeterias and restaurants.
5. External environmental factors (weather, vehicular traffic)
6. Failures in technological infrastructure (mobile networks, servers, real GPS)
7. Malicious behaviors or systematic fraud by users.
8. Offline social interactions (coordination through WhatsApp, informal groups, etc.).
9. Changes in market prices or economic aspects external to the system.
10. Large-scale logistics or motorized transportation outside the pedestrian/nearby 3 km radius.

Main Assumptions

- Agents act autonomously and follow rational decision rules with stochastic components calibrated with realistic data.
- Geolocation is accurate and always available within the 3 km radius.
- The system operates exclusively under the temporal restrictions of the project (same-day publication 2-4 hours before closing and collection within 1-3 hours).
- There is no massive bypass of the system in the base scenarios (although it is modeled as a risk in sensitivity scenarios).
- The agent population is closed during each simulation run (there is no dynamic user entry/exit within the same day).
- Reliability scoring directly influences the probability of receiving future assignments.

Parameters and Data

Spatial and Temporal Parameters

- Maximum radius: 3 km.

- Publication window: 2-4 hours before closing.
- Collection window: 1-3 hours.
- Temporal resolution: 10-minute steps.

Process and Resource Parameters

- Number of donors (cafeterias/restaurants): 5-20.
- Average surplus quantity per publication: 5-15 kg.
- Beneficiary capacity: variable by scenario.
- Base priority rate for verified charities: 20-30% scoring weight.

Stochastic Parameters

- No-show probability: 10-25% (normal or beta distribution).
- Notification response rate: decreasing with distance and time.
- Reliability score update: +1 for successful pickup, -2 for no-show.
- Variability in daily adoption.

Experimental Design

Main Scenarios

1. **Low adoption** (few initial users).
2. **Medium adoption** (realistic university scenario, ~70 kg/day recovered according to previous simulation).
3. **High adoption / Demand peak** (high concurrency, resilience testing).
4. **Scenario with mitigations disabled** (baseline to compare the impact of scoring and staged notifications).
5. **Sensitivity scenario** (±20% variation in no-show rates or charity priority).

Each scenario is executed with multiple repetitions (≥30) to obtain robust statistics.

Output Metrics and Validation Criteria

Key Metrics

- Surplus recovery rate (% kg collected vs published).
- Pickup completion rate.
- Fairness index (distribution to charities vs individuals).
- Average no-show rate and evolution of reliability scoring.
- Average time until assignment.
- Number of cascading reassignments.
- Environmental metrics (kg CO_2 equivalent avoided).

Validation Criteria

- Comparison against project success thresholds (≥80% recovery, ≥85% completion).
- Face validation with experts/stakeholders.
- Calibration and sensitivity: results must be consistent with previous process simulations and empirical data.

Traceability Matrix

Req / Risk ID	Description (Workshop 1 / 2)	Mechanism in Simulation (ABM)	Validation Metric	Status
REQ-01	Strict 3 km spatial restriction around the Faculty of Engineering	Euclidean / geo-referenced distance calculation between agents. Simulated `ST_DWithin` filter.	% of assignments within the 3 km radius (goal ≥ 99%)	Validated
REQ-02	Same-day temporal windows (publication 2-	Simulation clock control with configurable	Compliance with publication and collection windows	Validated

Req / Risk ID	Description (Workshop 1 / 2)	Mechanism in Simulation (ABM)	Validation Metric	Status
	4h before closing)	temporal windows.		
REQ-03	1-3 hour collection window	Timers per agent and surplus	% of pickups completed within the window	Validated
REQ-04	Mandatory priority for verified organizations (charities)	20-30% weight in the multi-criteria scoring of the matching algorithm	Fairness index (percentage of assignments to charities)	Validated
REQ-05	Matching based on proximity + reliability + priority	Serialized composite scoring algorithm implemented in the model matching engine	Average matching time and average score	Validated
REQ-06	Real-time push notifications	Staged notification model (500m → 1km → 3km) with variable delays	Response rate by notification level	Validated
REQ-07	Mitigation of no-shows through reliability scoring	Dynamic score update (+1 success, -2 no-show). Influence on future assignment probability	Evolution of average score and no-show rate	Validated
REQ-08	Automatic reassignment of uncollected surpluses	Behavior rule of the "Surplus" agent that launches a re-	Number of reassignments and time until new assignment	Validated

Req / Risk ID	Description (Workshop 1 / 2)	Mechanism in Simulation (ABM)	Validation Metric	Status
		matching event after timeout		
REQ-09	Historical record and impact metrics	Analytics agents that register all transactions	kg recovered, CO_2 avoided, recovery rate	Validated
REQ-10	Authentication and user roles (Donor, Beneficiary, Charity, Admin)	Role attributes in each agent with specific behavioral rules	Correct application of rules by role	Validated
REQ-11	Accessible mobile interface (simulated)	Agent decisions based on simplified interface (calibrated probabilities)	Participation rate by user type	Validated
REQ-12	Scalability under concurrency	Simulation of multiple simultaneous agents with matching queues	Performance under high load (demand peak)	Validated
REQ-13	Social fairness in distribution	Monitoring of Gini or fairness index in assignments	Fairness coefficient (goal ≥ 0.75)	Validated
REQ-14	Automatic generation of environmental impact metrics	Calculation of CO_2 equivalent per recovered kg	kg CO_2 avoided per day / year	Validated
REQ-15	Compliance with operational limits (no transportation or payment)	Explicit exclusion in agent rules	Non-occurrence of out-of-scope actions	Validated

Req / Risk ID	Description (Workshop 1 / 2)	Mechanism in Simulation (ABM)	Validation Metric	Status
REQ-16	Complete event logging	Detailed logging of all events and agent states	Log completeness (>98%)	Validated
REQ-17	Ease of maintenance and modularity	Modular design of agent classes and separated rules	Rule modification time (measured in experiments)	Validat

Critical Risks

Risk ID	Risk Description	Mitigation Mechanism in Simulation	Validation Metric	Status
R1	High no-show rate	Reliability scoring + penalty and reassignment	Reduction of no-show rate ≥ 50%	Validated
R2	Demand saturation / amplified demand	Staged notifications with delays	Fairness maintained under high concurrency	Validated
R3	System bypass (informal coordination)	Bypass model with low probability + penalty	System usage rate vs bypass	Validated
R4	Geolocation error	Configurable precision + noise scenarios	Robustness against position error	Validated
R5	Low initial adoption	Progressive adoption scenarios + score feedback	Adoption curve	Validated
R6	Imbalance between donors and beneficiaries	Dynamic adjustment of agents	Supply-demand balance	Validated

Risk ID	Risk Description	Mitigation Mechanism in Simulation	Validation Metric	Status
R7	Scoring degradation over time	Decay and recovery rules	Long-term score stability	Validated
R8	Notification failure	Simulation of partial failures and retries	Successful delivery rate	Validated
R9	Inequity toward charities	Fixed priority weight in scoring	% of assignments to charities	Validated
R10	Scalability under peak load	Stress simulation with 5x users	Maintenance of performance thresholds	Validated